CHAPTER

Data Life Cycle Processes

11

"In measuring a circle, one begins anywhere."

Charles Fort

Introduction

A life cycle is a series of stages through which an organism progresses from its conception to its death. Each stage is marked by a new level of development as an organism changes from an egg or seed to an adult able to produce offspring and then eventually dies. Some life cycles are complex (caterpillars completely change form when they become butterflies), while others are simpler and involve only growth and development rather than metamorphosis (a puppy looks like a small dog, with disproportionately large eyes). The life cycle phase and the characteristics of each phase for a particular species provide insight into the similarities among individuals of that species and the species' differences from other species. This is probably why the life cycle is a useful metaphor to describe other things, such as the conception, development, maturity, and obsolescence of products and the process of creating and managing data to realize value from it within an organization.

This chapter will examine several organizational processes that depend on the life cycle metaphor—the asset life cycle, the supply chain, the value chain, and the systems development life cycle (SDLC)—and show how they can be applied to data. The goal of the chapter is to demonstrate how knowledge of these cycles can be applied to build data quality into their supporting processes. My goal in presenting these cycles is to show how they inform each other. Multiple models show multiple perspectives and point to the reasons for complexity in a system like an organization. But just like a diagram of body systems (see Fig. 1), models also simplify complex systems and allow us to actually seem more than we can when we are trying to comprehend everything at once (Page, 2018).

The Data Life Cycle and the Asset/Resource Life Cycle

The data life cycle strongly resembles Juran's quality trilogy (planning, design, control) and the product life cycle that is the basis for it. As discussed in Chapter 5, data is a product of the processes that create,

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collect, and organize it. Life cycle management is an extension of the idea of quality control to all aspects of creating a product. Juran's *Quality by Design: The New Steps for Planning Quality into Goods and Services* is an extended explanation of how this works for physical products and services (Juran, 1992).[1]

Data can also be viewed as an asset that brings value through use, or, more generally, as a resource used to accomplish organizational goals. It has characteristics of both of these as well as some qualities that are distinctive to itself. If an organization is looking to improve overall data management and implement data quality by design in order to get more value from its data as an asset, then it helps to look at data from a higher level of abstraction: the data life cycle.

The phases of the data life cycle resemble those of other resources, such as equipment, finances, or even people (English, 1999). Acquiring and using resources requires planning. These resources then must be created, purchased, or hired. Often, work is required to prepare the asset to be used. People must be trained. Equipment must be set up and installed. To bring value, resources must be used. As English points out, "Idle resources produce negative value" (English, 1999, p. 201), meaning, of course, that they cost money. Many require maintenance, ongoing management, or tending to while they are being used. And, at the end of their useful lives, they must be appropriately disposed of; for people, we hope this means a happy retirement. Certain data must be destroyed when it is no longer legal or feasible to retain it. Economic value comes when the benefit from a resource's application is greater than the costs incurred from its planning, acquisition, maintenance, and disposition (English, 1999).

In addition to these standard phases of planning/preparation, creation, maintenance, use, and disposal, data also requires a level of design and enablement. It must be understood and its meaning documented if it is to be used by new people over time.[2] It also must be stored in a system where it can be both understood and accessed. The uses of data often result in the discovery of data issues (obstacles to the use of data), the creation of new data, or the identification of new data requirements. These situations mean there is a need to remediate or enhance data or to improve the processes by which it is created, maintained, stored, and shared. Data enhancement starts the life cycle again because enhancements require planning and preparation as well as data design and creation.

[1]This section relies on and draws significantly from McGilvray's *Executing Data Quality Projects: Ten Steps to Quality Data and Trusted Information* (McGilvray, 2021) and on English's *Improving Data Warehouse and Business Information Quality: Methods for Reducing Costs and Increasing Profits* (English, 1999). McGilvray's formulation of the data life cycle, POSMAD (Plan, Obtain, Store and Share, Maintain, Apply, Dispose) draws on English's work and expands it in significant ways. POSMAD describes and depicts the alignment of data with the universal resource life cycle. Using POSMAD as the basis for a life cycle model, I have added phases that are particular to data—Design, Enable, and Enhance—in part to more closely align it with Juran's trilogy. Like McGilvray's Store and Share, which recognizes that data must be made accessible to data consumers, these additional phases acknowledge conditions that are especially important for data. Especially when compiled from multiple sources, data is rarely in the exact form we need it to be in. It requires a level of design, along with the capture of explicit knowledge. It also must be set up in a system that makes it accessible. Loshin also describes an information life cycle: Create, Distribute, Access, Update, Retire (Loshin, 2011).

[2]See Joshik (n.d.), who points out that because "Many organizations spend 60 to 70 percent of their IS budgets on systems maintenance. . . . it is desirable that the documentation necessary for system operation and maintenance be produced as the by-product of the development process." I wish he had said "product," but I think the intention is there. Joshik includes data flow diagrams and data dictionaries as part of the documentation that should be produced by the SDLC.

Data life cycle phases include the following:

- **Plan/prepare:** Planning for data includes planning for each phase of the life cycle. Determine what data is needed as well as how it will be created, sourced, or acquired. Planning for known uses and ongoing support of use (e.g., training, customer support), access and security requirements, maintenance requirements, and archiving and disposal requirements. Planning should also include understanding and documenting the costs and anticipated value of the data.
- **Create/obtain:** Get the required data, either by obtaining it through an existing process, purchasing it, or establishing a process to create it.
- **Design and enable:** Define and design the technical and business processes to ensure that data can be accessed and kept current and complete. If there is a need to create new data, then design will be part of the create/obtain phase. If an organization will use data produced through operational processes, then design will be a separate phase.
- **Store and share:** Put data into a system that is secure and accessible to those who are authorized to see it and use it.
- **Maintain:** Implement operational processes that keep data current and complete and the monitoring functions that will alert production support teams and data users when unexpected conditions are detected.
- **Use/apply:** Use data in operational processes and reporting, analyze data to identify trends and business opportunities, incorporate the data into models and other applications, and, potentially, sell the data as a product.
- **Enhance and improve:** Recognize gaps and potential improvements, report issues and unexpected behavior in the data, and formulate proposals for improving data content and structure.
- **Dispose of:** Ensure that data is removed from systems and processes when it has reached the end of its useful life. Archive data that must be kept for legal or other reasons. Dispose of data that must be destroyed. Disposal schedules may be driven by regulation, contractual obligations, business requirements, or a combination of these things.

Having a picture of the data life cycle allows us to see that the activities required to manage data and ensure its quality differ from phase to phase (Fig. 39). In terms of quality, the most important points in the data life cycle are when data is created and when data is used or applied.[3] Creation provides the best and most obvious opportunity to ensure that data is complete and accurate. However, quality characteristics must also be defined when data is enhanced as this is a new moment of creation. In addition, as will become clear when we look at the data supply chain, data is often changed as it moves through the organization. A “moment of creation” can occur multiple times within the supply chain if data is aggregated, calculated, or transformed. The only time data brings value is when it is used, and if it is not of high quality at that point, its use may result in incorrect conclusions, bad decisions, or negative interactions with customers or other stakeholders. In such cases, data ceases to be an asset and functions as a liability. An organization actually loses value if its data leads to mistakes that negatively affect its stakeholders.

[3]This insight comes from Tom Redman, who discusses the importance of data creators and data customers in the overall management of data quality, and from Danette McGilvray, who observes that although all phases of the life cycle involve cost, benefit is created only when data is used (McGilvray, 2021, p. 38).

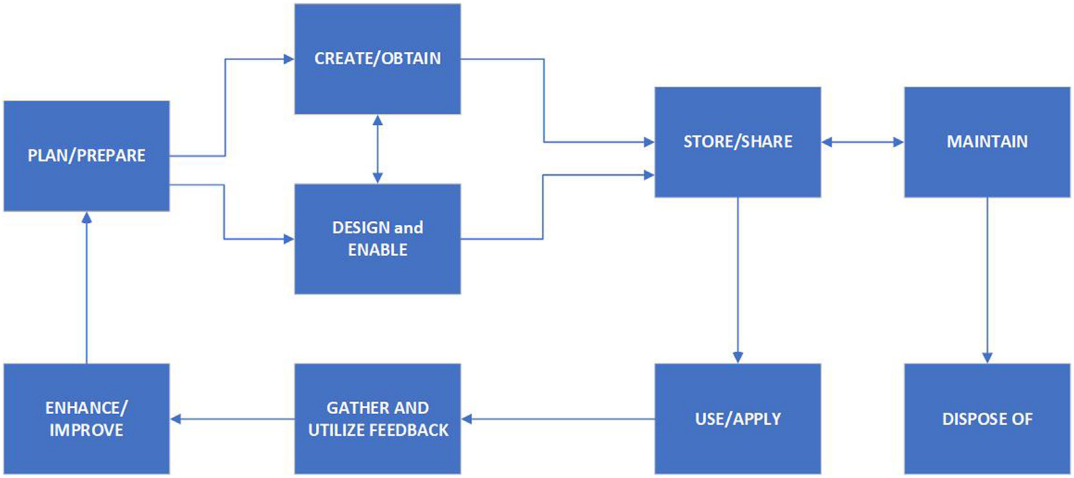

FIGURE 39

Data resource life cycle.

Managing Quality Throughout the Data Life Cycle

Although creation and use are the most important points in the data life cycle, the quality of data can be influenced at any point in the life cycle. This is related to the fact that, while we refer to a data "life" cycle and understand that certain characteristics of data are similar to organic materials (e.g., grows over time, takes on new meaning, "reproduces itself"), data is not actually organic. It is created by people or by machines designed by people, which means we have a lot more control over its creation than we give ourselves credit for. It is worth taking another look at the life cycle phases with this observation in mind. Doing so allows us to see how quality can be managed throughout the data life cycle.

Plan/Prepare

Planning includes defining expectations and requirements for quality and putting in motion the activities required to use the data over time.

- **Clear criteria for quality:** If expectations for data quality are unclear or undefined, then data will not be created consistently or correctly. If controls are not put in place to reduce the chance of errors, then more errors will occur. If processes do not account for how data will change over time and maintenance is not built in, then quality will deteriorate over time. If data creators do not know how the data is used and who uses it, then they will not understand the purpose or importance of the data they create, and quality will deteriorate over time. For these reasons, planning for data should include explicitly defining requirements for quality. These may be based on data standards, dimensions of quality, process requirements, data consumer requirements, or a combination of these things.

- **Supported by metadata:** Planning and preparing must account not only for data, but also for the information required to use the data—the metadata. Documented knowledge about data is essentially for its use. Without comprehensible definitions of data sets and data elements and without information about the processes that create data, data consumers will perceive data to be of low quality, regardless of the data's intrinsic "correctness." The limits of their own understanding will influence the level of trust they have in the data.
- **Account for the costs of poor planning:** The need for metadata increases as the ability of organizations to produce and use data increases. There is simply less time to learn, so if data is not documented, knowledge about it disappears into the ether. In the absence of usable definitions, models, specifications, and data flow diagrams, every data consumer must recreate the knowledge on his or her own. This extra work, repeated multiple times, diminishes the potential value an organization can get from its data. It increases costs (people must recreate knowledge about data) and risks (in recreating this knowledge, they will make mistakes and create discrepancies) while reducing the time spent using and thus deriving benefit from data.

Create/Obtain

Data can be obtained from an existing process or establishing a process through which it can be created. In organizations focused on using analytics to improve their performance, the most important source of data will be its own operational or transactional processes. Many organizations will augment this internal data with data from external sources that may provide, for example, additional demographic information about customers. In such cases, obtaining data requires finding a source or broker who can supply it. Because its quality will depend on the supplier, it is important to have explicit quality requirements when selecting a supplier.

Except for organizations that directly monetize data, it would be rare to create a new process simply to produce data. However, when new products or services are introduced, it makes sense at the beginning of the process to account not only for the data that will be created but for how data will be collected as these decisions have a direct impact on data quality. The create/obtain phase should include documenting the following:

- **Clear criteria for quality:** In creating a new process through which to obtain data, an organization has the opportunity to build quality into the process itself. This includes defining quality expectations up front and defining controls within the process to ensure that these expectations are met.
- **Knowledge of the creation process:** When obtaining data from an existing process, start with the assumption that the data may not yet be fit for purpose. Understand how that process works, including the conditions under which data is created and the incentives for particular outcomes. For example, if a claim adjudication process is focused on the speed at which claims are processed, it will produce one type of outcome; if it is focused on the accuracy of claims or on reducing the risk of reversals, it will produce another type of outcome. If it is focused on the accuracy of clinical information rather than the accuracy of contractual information, it may produce yet a different outcome. Understanding the purpose and drivers of the creation process will guide the creation of controls or other requirements.
- **Specifications for data suppliers:** When purchasing data, define quality requirements. Even if these cannot be met, having them documented is a vital step in selecting a supplier. It will also allow for the identification of risks and gaps.

Design and Enable

Design and enablement includes planning for the business and technical processes that ensure quality. These can take place at the application level to support the execution of the process, or they can take place at the level of the data store to support analytics and reporting. Design must account for both the technology through which data is created or shared and the business processes the data supports. Design directly affects usability, which affects both measurable quality characteristics and data consumers' perception of quality.

- **Enforce quality:** Designing an application to support a business process that produces high-quality data requires understanding the desired outcomes and putting in place the technical controls and validations that support those outcomes. Design can help prevent errors, for example, by restricting a domain of possible values (through drop-down lists or lookups), implementing logic based on business rules (birth date cannot be a future date), or preventing a process from moving forward if input data represents an invalid condition (a Zip code that is not correctly associated with a city; an invalid credit card number).
- **Enforce an enterprise perspective:** Enabling data for storage and use in analytics includes understanding how data will be brought together, often from different sources, and resolving problems of integration (granularity, differences in meaning, gaps at the attribute level across different data sets) and orchestration (timing and coordination of data delivery and processing) before they happen. An enterprise perspective includes setting standards for architecture, modeling, and development work to ensure that data is created, stored, and maintained consistently across applications. These processes can have a large impact on the quality of data. The enterprise perspective should include system controls to ensure that data sets are complete as they are brought into a warehouse, mart, or other storage platform. It should also include data quality monitoring routines that validate data against rules and patterns, including measuring referential integrity.

Store and Share

Storing and sharing are the equivalent of a physical distribution method, but for data. They are the means by which authorized data consumers access data and put it to use. How data is stored can have a direct impact on quality because it has a direct impact on the people and processes that access data. The ability to share data affects the value an organization can realize from data because it can influence the work associated with data use.

- **Standardize data storage:** If data across the enterprise is created based on consistent standards (ideally, through an enterprise data model), it is easier to store and share. If data is not created consistently, it should be standardized as part of data storage to make it easier to use across the enterprise. If data is difficult to access or share, then it will require additional work to use it, thus reducing its value to the organization.
- **Enable appropriate access/prevent inappropriate access:** Data brings value only when it is used, so enabling people to access data is critical to an organization's ability to leverage data. At the same time, inappropriate access to data represents risk. Planning for data storage should

also include planning for access controls that prevent data misuse by internal data consumers, and data security architecture that prevents access by external agents.

- **Support data sharing through appropriate metadata:** All stored data should be supported by metadata that enables it to be understood and efficiently accessed. This metadata should include information about data quality conditions, measurements, and issues.
- **Help data consumers understand data structure:** The way that data is physically stored has an impact on how it is accessed and shared. Few data consumers will understand the finer points of data modeling or physical storage, and they do not have to. But they will need foundational metadata to find and query the data they want to use.
- **Recognize the impact of access tools on the perception of data quality:** The sharing part of store and share is also important. Different access tools can affect the quality of data. For example, different tools may present numeric data at different levels or precision or have different methods for rounding numbers. These differences can influence people's perception of data quality, especially when different tools that make the same data set appear to have different characteristics.

Maintain

Maintenance processes also influence quality. Data currency is directly dependent on the method and frequency through which data is updated. Some maintenance processes may make data less usable. For example, fully refreshing a data set when a data consumer requires detailed history will make data unfit for purpose. The history will be lost. Similarly, data with complex history can be hard to use if a data consumer must focus only on the most current version of the data set.

- **Define maintenance requirements as part of quality requirements:** Data consumers will want to understand how historical data is maintained: are existing rows updated or fully replaced, or are new rows added with updated information? They will need to know what actions result in a row being updated versus a new row being created. Also, they will want to know the timing between when data is created via its originating process versus when it will be available for use.
- **Include metadata about maintenance processes:** Among the metadata required to support the use of stored data is information about its maintenance processes, including the maintenance schedule and the approach by which data is maintained (e.g., fully refreshed versus incrementally updated, versus a combination). These details enable data consumers to understand whether data is current enough to meet their needs as well as whether data contains the level of historical detail required for their analyses. Stored data should also include audit columns that timestamp when rows were created and updated.

Use/Apply

As noted earlier, the only phase in the life cycle during which an organization gets value from data is when data is used. All other phases should support the use of data. Data use is also the most important source of feedback about data quality because data consumers are the customers of the processes that create and manage data.

- **Support data use with metadata:** Using data requires knowledge of what the data represents and how it represents its concepts (e.g., knowledge of the process by which it is created and often knowledge of the system in which the data is stored and maintained). This knowledge comes in the form of metadata—documented information about the meaning and structure of the data. If metadata is not stored with the data or made accessible to enable use and sharing, then the costs of usage increase, diminishing the value that can be gained from data. The level of knowledge needed to use data corresponds to the usage task at hand. A person who is running a canned report needs a lot less data knowledge than a person creating a new report, using data in reporting, or analyzing data to identify trends and business opportunities.
- **Improve metadata quality through data use:** As noted under "Plan/Prepare," managing the data life cycle also includes managing the metadata life cycle. Metadata required for data use should be created/obtained as part of the development process. It should also be maintained as data uses evolve. Metadata improvement can include information about the ways data can be accessed and used as well as on the quality expectations related to the data.

Gather and Utilize Feedback

Feedback is critical to the improvement of any system, especially a system for managing data. Because the goal of data is to represent people, objects, events, and concepts in the real world, data that is disconnected from the reality it represents quickly deteriorates in quality. Systems theorist Ken Orr put this very directly:

> The principal role of most information systems is to present views of the real-world so that people in the organization can create products or make decisions. If those views do not agree with the real-world for any extended period of time, then the system is a poor one, and, ultimately, like a delusional psychotic, the organization will begin to act irrationally (Orr, 1996, p. 2).

The data life cycle can be viewed as a system for managing data. Its subcomponents include the business processes and systems through which data is created, enabled, maintained, stored, accessed, and used. Such a system is likely to generate feedback about the meaning of the data, its quality, and the ability of data consumers to use the data. Feedback represents the opportunity for improvement of the following:

- **The meaning of the data:** Questions about the meaning of the data represent opportunities to improve metadata.
- **The quality of the data:** Feedback about the quality of the data may also represent an opportunity to improve metadata. Such feedback may identify issues that require remediation, requirements for new data, or other improvement opportunities.
- **The ability to use the data:** Feedback on data use may reflect limitations of data structure, tools, or the data itself.

Correct, Enhance, and Improve

When people use data, they see gaps and issues and identify opportunities for enhancement or other improvements. Even if individual data consumers do not propose changes, business processes and

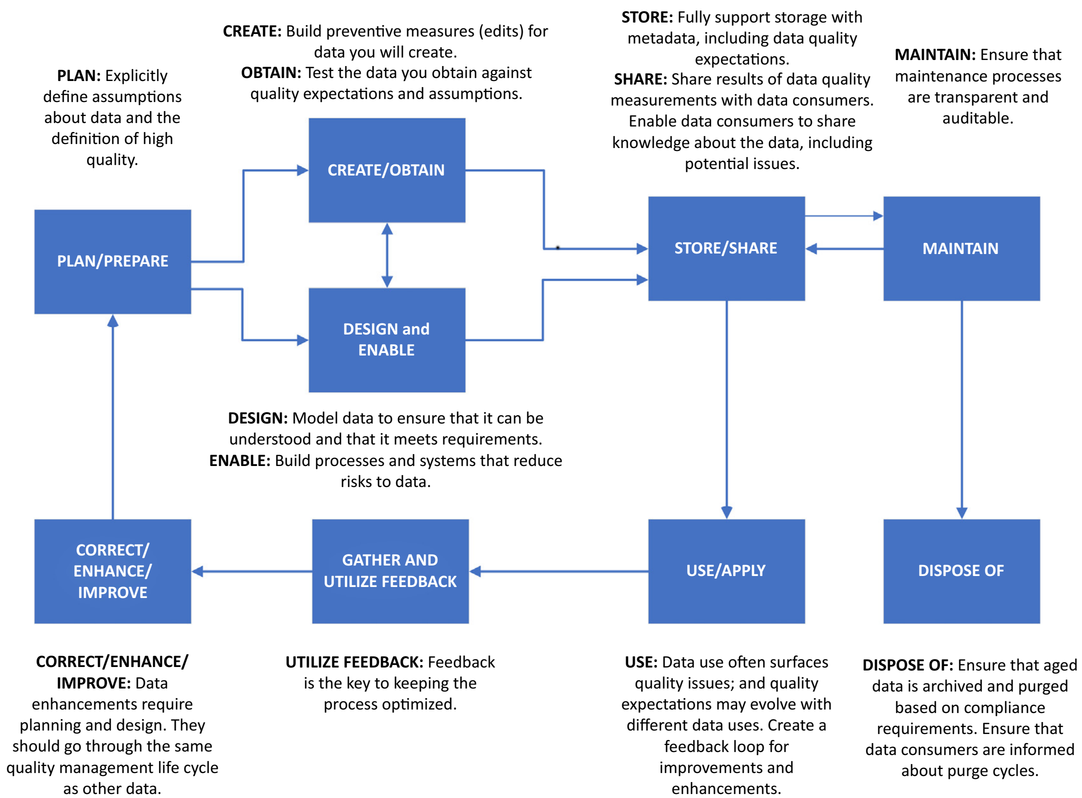

FIGURE 40

Data life cycle annotated with data quality considerations.

compliance requirements evolve over time. Data creation processes and data storage applications change over time through remediation and enhancement. The process of improving data restarts the life cycle process (Fig. 40). Improvements require planning, obtaining data, and so on. They require evolving metadata along with the data itself.

Dispose Of

When data reaches the end of its life cycle, it must be properly disposed of. While, in theory, data could last forever, in practice, a lot of data is not allowed to last forever. For some data, retention periods are regulated, and organizations are required to purge data from their systems and securely dispose of it. Likewise, for legal or regulatory reasons, organizations may be required to keep other data for a set period of time, even after the organization itself no longer uses the data. Disposal is often a two-step process that includes first archiving data (storing it in a largely inaccessible environment) and then actually destroying it. Managing the retention and disposal of data is part of overall life cycle management and should be accounted for up front as part of planning because data at different phases of its life cycle may be stored and managed differently. These differences can affect the costs of storing, sharing, and accessing data. Planning for storage and retention should include accounting for audit fields (e.g., record status codes, create and update dates, record begin and end dates, user IDs, and process IDs) that might be required to ensure data meets archiving requirements.

Benefits of Understanding the Data Life Cycle

There are multiple benefits to understanding the data life cycle. Obviously, doing so allows an organization to plan for the overall management of the data and its associated metadata. This planning has direct effects on the organization's ability to manage the quality of data as well as to meet legal and regulatory requirements. In addition, as will be discussed in the next section, understanding the life cycle of individual data sets also enables the organization to develop a clearer picture of the data supply chain, and from that its value chain. Knowledge of the data life cycle enables an organization to evaluate the costs and benefits associated with a data set during its lifetime. If, for example, an organization wants to understand whether it is getting value from its customer data, it can determine the costs of obtaining, storing, and maintaining customer data against the value brought by using that data. Knowledge of these costs and benefits can help answer questions about how long the data should be retained before it is disposed of and whether there is value in enhancing the data.

The Data Supply Chain: Moving Data Into and Within an Organization

The data life cycle focuses on how different data sets will be managed over time to bring value to an organization, but it is a relatively abstract, system-agnostic concept. The data supply chain focuses on the details of how particular data moves into and through the processes and systems in an organization. The supply chain highlights the relationships among functions that create and use data. In doing so, it allows the organization to see where data is expected to fit together and interact with other data. It allows planners to identify data quality requirements related to data exchange and interoperability. This knowledge, important for internal processes, is even more critical for organizations that obtain data from external entities (hence ISO's definition of quality data as "portable" data).

People who are familiar with data management terminology may be asking, "Aren't you talking about data lineage?" I would respond, "Yes and no." The concept of lineage—descent from a common ancestor—provides an important lens for understanding data within an organization. Data lineage is a form of metadata that describes where data originates, how it moves through an organization's systems and processes, and, importantly, how it is transformed as it moves. As described in Chapter 10, lineage can be depicted at a high level (architectural diagrams are a form of data lineage), or it can be very detailed (as code that describes the input to and output from specific transformation rules under specific conditions). A documented data supply chain diagram is a form of lineage, but not the only form.

The data supply chain is focused on who creates and uses data and for what purposes. Thus, it provides an excellent pathway into discussions about data quality expectations as well as about the costs and benefits of the organization of the chain. Documented knowledge of both lineage and the data supply chain are also critical when data quality issues arise. They provide important input to root cause analysis as well as the business impact of data issues.

Supply Chain Management Defined

A simple definition of the supply chain in manufacturing is the set of processes involved in the production and distribution of a product or commodity. A more elaborate definition acknowledges the complexity involved with getting a product to market:

> A supply chain is the network of all the individuals, organizations, resources, activities and technology involved in the creation and sale of a product, from the delivery of source materials from the supplier to the manufacturer, through to its eventual delivery to the end user. The supply chain segment involved with getting the finished product from the manufacturer to the consumer is known as the distribution channel (Tech Target, n.d.).

Supply chain management is oversight, coordination, and integration of the flow of materials, information, and finances as they move in a process from supplier to manufacturer to wholesaler to retailer to consumer (Tech Target, n.d.).

A well-managed supply chain enables efficiencies. Manufacturers create and ship only as much product as can be sold. This helps suppliers, manufacturers, and retailers manage inventory. Supply chain management also helps build working relationships because managing a supply chain requires setting clear expectations related to timing, quality, and communications necessary to delivering products on time, to specification, and to the right locations (Perkins & Wailgum, 2017).

Data Movement as a Supply Chain

The concept of a supply chain provides a useful metaphor for how data moves within an organization. It acknowledges that there is a beginning point to data (data provenance)—its creation or purchase—and that data can move through a series of processes and activities before it is used by a data consumer. The idea of a "chain" helpfully implies that these processes are linked together and dependent on one another. It also reminds us that a weak link endangers the whole chain.

The recognition that a supply chain includes not only the flow of materials, but also of finances and information is instructive. Data is not free. It costs something to create data or bring data into an organization and to share it. Information—various forms of metadata, a knowledge of technical and business processes, an architectural approach—is required to move data and, ultimately, to enable people and processes to use data. Because in many organizations data producers may not be aware of their customers, internal data consumers, the supply chain model can raise awareness of how data moves within an organization, and of the resources and channels required to move data from its origin to its multiple uses.

Depicting this chain also provides a way to rationalize and simplify data movement. Anyone who has ever seen a "spaghetti" chart of an organization's systems architecture will admit that in many organizations, data is moved among systems more often and in more complex ways than most of us would guess. Having a visual of these relationships is a first step to reducing this complexity.

Fig. 41 presents a very simplified diagram of a data supply chain for a health care insurance company. Data about clients (commercial entities that provide health care benefits to their employees), members (the employees who receive these benefits), and products (the benefits themselves) all combine to identify who is eligible for which benefits at which time. Data about health care providers, their contracts, and the networks they are part of is used in combination with eligibility data to pay claims. Claim experience of a given client is used by underwriting to determine the premium the clients must pay. The clinical results of treatments are used to rate providers. These ratings influence provider contracts. Much of this data then is used in financial reporting. This simple depiction of a supply chain nevertheless shows that there are complex relationships between data producers and data consumers, with most systems and processes playing both roles.

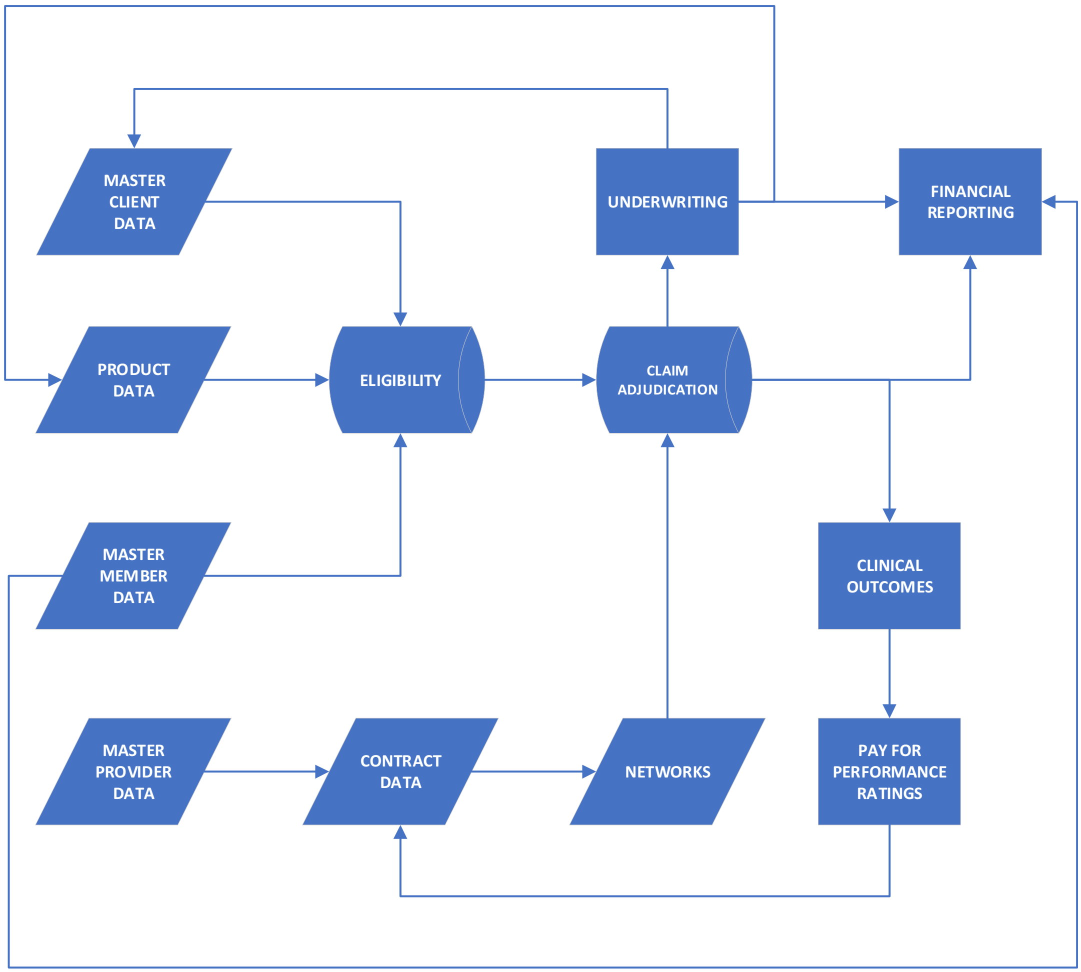

FIGURE 41

Conceptual data supply chain for a health care insurance company.

The Supplier/Purchaser Relationship

In manufacturing, successful supply chain management requires a mutually beneficial relationship between suppliers and purchasers. This mutual benefit depends on clear expectations, including expectations for quality. If a purchaser cannot meet a price, the supplier does not deliver the goods. If a supplier cannot meet expectations for quality or timely delivery, a purchaser can go elsewhere for materials. If the parts they supply do not meet specifications for the inputs, suppliers can, in some cases, be held responsible for the quality of finished products.

A successful supply chain depends on a level of balance and equality. If there is a lack of balance, the process may not support quality outcomes. For example, if there is only one supplier for a given input, then the purchaser is put in a position of having to accept the inputs regardless of timing or quality. If there is only a single purchaser for an input or if the market is dominated by a very large purchaser, then suppliers may not get the price they need for their goods and may be forced out of business. Manufacturers must manage their supply chains to make a profit.

Organizations may not even think about their data as part of a supply chain. The first step to improving the data supply chain is recognizing that it exists. The next steps are assessing how well it works and identifying opportunities to make it work more smoothly. In looking at their data supply chain for the first time, organizations are likely to recognize some common characteristics:

- **Data "suppliers" do not always think of themselves that way**. They see their data more as a by-product of their operational processes than a product to be used by others in the organization. Because of this, they are not likely to think of people and processes who use the data as "customers" whom they are trying to please. Yet, case studies show that when data suppliers become aware of their customers, even minor changes can help improve data quality (Kenett & Redman, 2019).
- **Internal data supply chains are often out of balance**. The option of finding a new source of data, a new supplier for organizational data, rarely exists. When a project requires membership data within a health care insurance company, the data must come from the eligibility system. There is no other option. However, once the people managing eligibility information understand their role as a supplier, again, even minor changes can help make their data more usable downstream.
- **Like data itself, the data supply chain has an organic nature**. It is not as linear as the manufacturing supply chain. In many organizations, data moves in patterns that more closely resemble a web or a data mesh (see Fig. 41). Once supplied, data can be used for new and evolving purposes, without the suppliers even being aware of these uses.

These risks, however, are among the most important reasons why it is useful to envision data moving in a supply chain. If people within an organization understand how and by whom data is created and used, then they will be better able to manage data. Knowledge of downstream data uses enables the establishment of clearer expectations. Knowledge of business processes that create data can be used to improve those processes and enable higher-quality data. Known gaps can be identified and, in some cases, filled if the data producers are aware of requirements. For example, if the quality of customer address data is poor, an organization can improve it by purchasing reliable address data from an external source—to which they can and should state clear data quality expectations.

Knowledge of the supply chain amounts to a specialized form of knowledge about the organization. The handoffs of data from one process or system to another will work better if the supplier understands the quality requirements of the process being supplied with data. Knowledge of the supply chain also introduces the opportunity to put in place controls to identify the extent to which quality expectations are met. Improving the supply chain starts with seeing it. The goal is not necessarily to make it perfect, but to clarify the connections and, where possible, to simplify it so as to reduce risks and redundancies.

The Value Chain: Finding Efficiencies and Adding Value

If the supply chain model allows us to picture how materials move through a set of processes to become products or services, the value chain allows us to evaluate the costs and benefits of the activities and purposes associated with this movement—a kind of lens through which to view supply chain activities. The value chain envisions an organization as a system (a set of things working together as part of a mechanism, network, or method, according to which something gets done) that uses resources (inputs) and transformation processes (subsystems) to create products and services (outputs) (Porter, 2008).[4] Resources include money, people, materials, equipment, information, and other processes (e.g., management, administration).

Each activity in the value chain has costs and may add value to the end-product. The overall value chain represents the costs of creating a product and shows where value is added to the product by each step of the process. Economic value is produced when the benefit from a resource's application is greater than the costs incurred from its planning, acquisition, maintenance, and disposition.

The primary activities in a value chain—those that add value directly—include the following:

- **Inbound logistics:** Activities required to receive, store, and disseminate inputs
- **Operations:** Activities required to transform inputs into outputs (products and services)
- **Outbound logistics:** Activities required to collect, store, and distribute products and services
- **Marketing and sales:** Activities required to inform potential customers about products and services
- **Service:** Activities required to keep the product or service working after purchase

These are supported by secondary activities, which add value indirectly (e.g., procurement, human resources management, technological development, and infrastructure)

Value chain analysis is a form of process analysis that considers each step in a value chain to determine whether it adds to competitive advantage. In doing so, the analysis can identify process points that may be disadvantageous or "low value-add." Value chain analysis creates knowledge about an organization's operational activities, including the cost drivers for each activity and the

[4]Porter's definition of a system is similar to Meadows's: a system is "a set of elements or parts that is coherently organized and interconnected in a pattern or structure that produces a characteristic set of behaviors often classified as its 'function' or 'purpose'" (Porter, 2008) and Deming's: a system is "a network of interdependent components that work together to try to accomplish the aim of the system. A system must have an aim. Without an aim there is no system" (Deming, 1994).

relationship among activities. Recognizing how activities depend on one another can help an organization identify opportunities for reducing both cost and risk. Value chain analysis can also be used to improve products by recognizing activities that create features that customers value most and finding opportunities for an organization to differentiate its products and services from those of its competitors.

From the perspective of data quality management, the value chain model implies a set of questions that can be asked about the processes that enable movement along the data supply chain.

- **Inbound logistics:** What costs are associated with creating or purchasing data and moving it from where it is created to where it will be used? What subprocesses can enable data to be better prepared so it is easier to use and can be used multiple times?
- **Operations:** What work is required to make data secure, accessible, current, and prepared for use? What processes are in place to enable data to be used multiple times? How is metadata maintained to simplify the use of data?
- **Outbound logistics:** What costs are associated with distributing data within the organization? Where are there redundancies in data storage? How is data reuse incentivized and data replication discouraged? How can access be simplified? How does the organization track the value of its uses of data?
- **Marketing and sales:** What are the costs of making data consumers aware of what data is available? What information is available to encourage data consumers to use data sources that are designated "authoritative sources," "systems of record," or "sources of truth"?
  - Note: Unfortunately, the marketing and sales aspects of the data value chain are often ignored, resulting in reduced value to the organization. Investing in making data consumers aware of existing sources of data can add value by reducing data redundancy. Redundant data reduces value directly (it costs time and money to create and store redundant data) and increases risk (redundant copies of data are more likely to get out of synch with their sources and with each other). Again, reliable metadata (e.g., a comprehensive data catalog) adds value and reduces risk.
- **Service:** What are the costs of maintaining data and supporting data use? How are data consumers informed about changes and updates to data? How are they informed about data quality levels? How are they made aware of issues that may affect their use of data?

The organic nature of data complicates the value chain and the supply chain. Neither one is linear. They can be hard to depict. Nevertheless, insight into the current state of the overall process can help identify ways to reduce waste and improve value.

The formal process of defining a data value chain can be accomplished through the use of information product maps that depict the transformation of a data unit into an information product that can be used by a data consumer (Fig. 42). The associated costs, value, and quality requirements can be mapped so the overall value of the data unit can be calculated (Heien, 2012). For example, if it costs $5 each to produce data in two sources and $5 to process the data, the total cost would be $15 for the information product. If the product is worth $10 each to the two consumers who use it, then its total value ($20) is greater than the cost of producing it, and the organization realizes $5 in added value (Fig. 43).

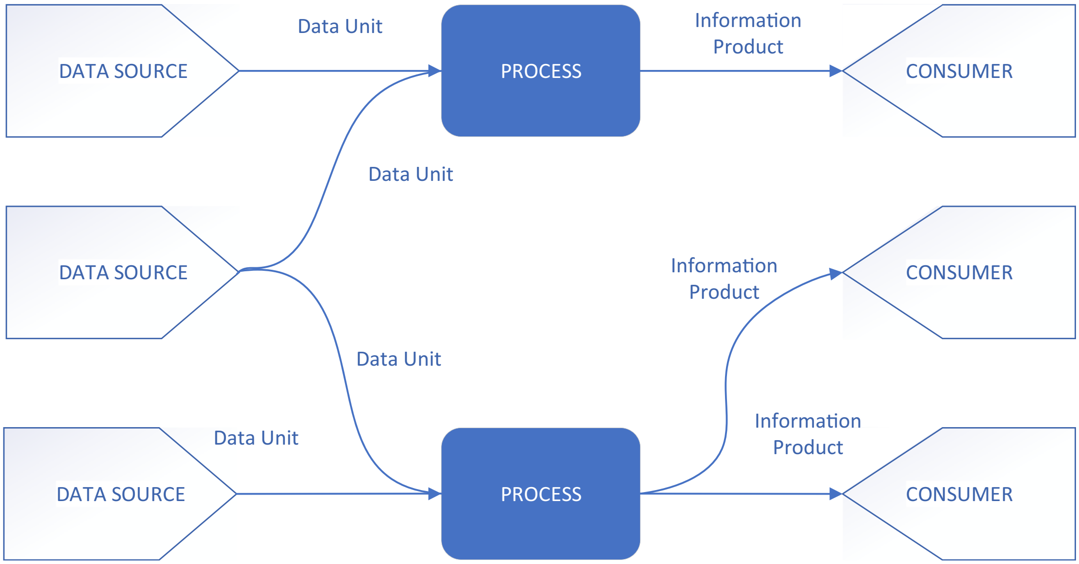

FIGURE 42

The basic components of information product maps.

From Heien, C. H. (2012). Modeling and Analysis of Information Product Maps (Ph.D. Thesis). University of Arkansas at Little Rock. Used with permission.

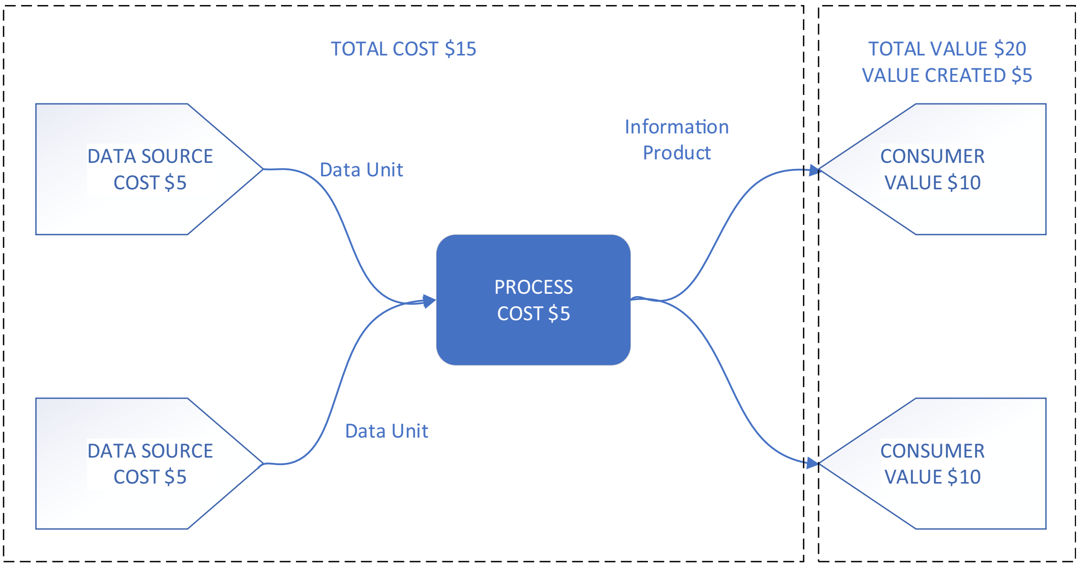

FIGURE 43

Basic usage of metadata in information product maps.

From Heien, C. H. (2012). Modeling and Analysis of Information Product Maps (Ph.D. Thesis). University of Arkansas at Little Rock. Used with permission.

These models are simplified to illustrate how cost and benefits, and thus, value, can be understood within the process of data creation and use. Such depictions have multiple uses:

- Mapping data sources to information products to consumers of those products raises awareness of the data connections within the organization and improves understanding of the data supply chain.
- This knowledge can be used directly and indirectly to improve data quality because stakeholders along the data chain can see their relationships and define their requirements for quality.
- Overlaying the costs of the processes and the value to data consumers allows for the calculation of value to the enterprise.
- Knowledge of how data is used can be applied to identify potential reuses of data to derive additional value.
- Analysis of information product maps can be extended over time and across data consumers to get a clearer picture of the value of data to the organization.

This knowledge can inform a range of processes, including identifying which information products to invest in and ways to direct data consumers to existing information products to get more value from them. This is because each additional use of the same information product creates more value at a significantly reduced cost. For example, in Fig. 43, if a third data consumer were to use the same information product and benefit to a value of $10, the organization would get a total benefit of $15, provided the costs are fixed.

Information product maps can also be used to understand the overall costs of managing a data set. As noted, value is the difference between the benefit gained through the use of an asset and the costs associated with all of the other life cycle phases. When the benefits outweigh the costs, value is created. When the costs outweigh the benefits, value is lost. If data is designed to be of high quality, then it is relatively easy to measure the costs associated with each life cycle phase (because there are not a lot of hidden costs), to determine the benefits gained by data use, and to compare the two to see how much value the data brings.

Because data can be used without being consumed, each additional use of data can add value, with limited additional cost. If an organization has a complete and current list of customers, with correct addresses, phone numbers, contact names, and email addresses, and if the process to maintain the data has reliable controls, the sales department can use it to contact customers, the marketing department can use it to communicate about new products, the shipping department can use it to ship products, and management can use it in reporting.

If data is not of high quality, then an organization will incur unexpected costs related to poor quality, for example, scrap and rework, combined with the risk and inefficiency of multiple users cleansing or preparing data that they thought was ready for use. These activities reduce the value by creating more cost. The costs themselves may be harder to measure because they are usually hidden. (No salesperson wants to tell their boss that they spent the day cleansing data, rather than connecting with customers.) But whatever they amount to, we know that with each use of the data, they will increase. Fig. 44 depicts this situation, showing the impact of data quality remediation on the value created in Fig. 43. Just as value increases with each use of a high-quality information product, cost and risk increases with each use of a poor-quality information product.

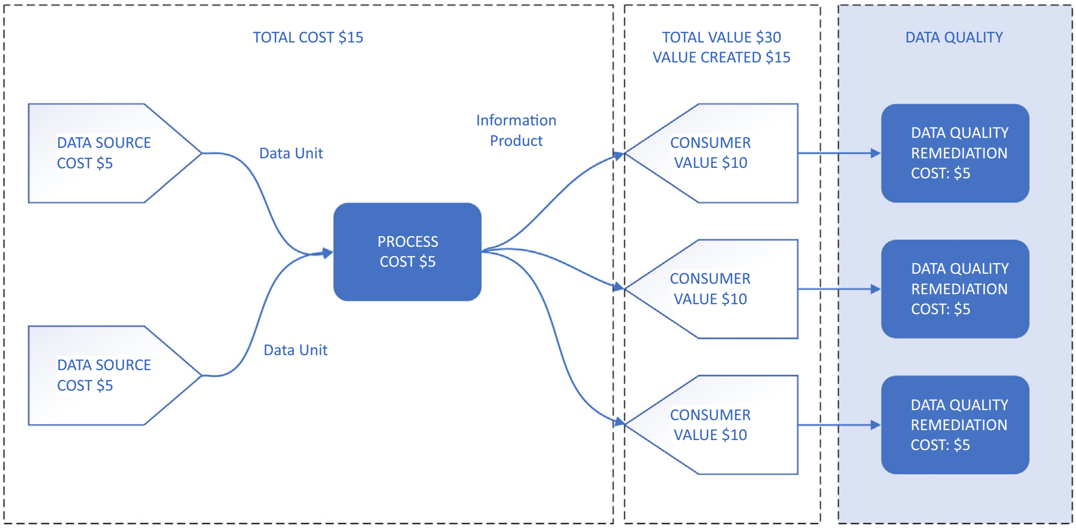

FIGURE 44

Poor-quality data reduces the value of information products.

Modified from Heien, C. H. (2012). Modeling and Analysis of Information Product Maps (Ph.D. Thesis). University of Arkansas at Little Rock. Used with permission.

These hidden costs add risk to the use of the data, which may further reduce its value. The sales department may change the data one way, while underwriting makes a different set of changes, and marketing makes yet a third set. Soon "the same" data set becomes multiple data sets that data consumers assume are "the same." If they are aware of these differences, they may pay to get the data "fixed." If they are not aware, they may use it and confuse their customers with conflicting information.

It costs organizations money to create and share data, but few organizations programmatically recognize the costs of creating data as such. And yet, the benefits of this kind of analysis are clear. It can identify opportunities for efficiency with data production, management, and distribution; encourage reuse; and fuel conversations about quality. A better understanding of the value chain can inform data strategy, solution design principles, and architecture standards. Better consumption and use models can be implemented for shared data assets.

The Systems Development Life Cycle

Most people will recognize a similarity among the asset management life cycle, the supply chain, and the SDLC.[5] Indeed, each describes a methodology for managing a complex process through

[5]The SDLC is also sometimes referred to as the *project life cycle* because the development phases must be managed to schedule and to budget. I have tried to focus on the system part of the concept, but I have slipped between the two on occasion.

well-defined phases. However, they focus on different things. The data life cycle is about planning for data to meet organizational needs and enabling the use of that data over time as data is stored, shared, maintained, and applied within an organization. The data supply chain describes the connections between data producers and data consumers and how they rely on each other. The value chain describes the costs and benefits of producing and using information products. The SDLC is about implementing technical functionality to meet the range of organizational goals that are dependent on data. That technical functionality enables the internal data supply chain. (It should be noted that the technologies implemented via the SDLC also have a life cycle outside of the projects through which they are implemented [DAMA, 2011; English, 1999]).

There are risks in confusing these, though, especially because most SDLC methodologies do not require that teams pay specific attention to the quality of the data itself. To the extent that SDLC methodologies recognize data as an input (and they do), they do not account well for the work of data quality assessment or for the possibility that available data may fail to meet either stated requirements or implicit expectations for quality.

A lot of work to ensure that an organization has high-quality data could be accomplished via information technology (IT) projects. It is true that some IT projects are purely technical, such as those that focus on implementing new infrastructure or improving the performance of a system or platform. However, most focus on one or a combination of the following:

- **Application development:** Creating or implementing new applications to execute operational business processes or support analytics
- **Application enhancement:** Improving the functionality of existing applications for operations or analytics
- **Data integration:** Integrating data from multiple sources into data stores for use in operations or analytics

These have a common thread: the goal, from a business perspective, is either to create or use data and information. These projects extend or enhance the data supply chain. But they rarely account for the data.

This point is often missing in discussions about IT projects. For example, when Wikipedia observes "a system can be composed of hardware only, software only, or a combination of both," someone seems to have forgotten that information systems contain, well, information. The Business Dictionary tries harder, defining an information system as "a combination of hardware, software, infrastructure and trained personnel organized to facilitate planning, control, coordination, and decision making in an organization." This includes a lot more than Wikipedia's hardware and software, but an essential ingredient—data or information itself—is still missing.

Fortunately, one does not have to go back too far to find a definition of information systems that emphasizes information: "An information system . . . [is] a set of interrelated components that collect (or retrieve), process, store, and distribute information to support decision making, coordination, and control in an organization" (Laudon & Laudon, 2001, p. 7). In this definition, an information system performs four activities that bear a very strong resemblance to Juran's definition of creating a product ("a product is the output from any process" [Juran, 1992]):

- **Input:** Collect or capture data from the organization or its environment.
- **Processing:** Organize it into a more meaningful form.

- **Output:** Transfer the processed information to the people who will use it to make decisions, control operations, analyze problems, or create new products and services.
- **Feedback:** Output that is used to evaluate or correct the input or processing stage.

An SDLC is a process for creating or enhancing an information system. Like an asset/resource life cycle, an SDLC provides a lens through which to think about data and data quality. If we can focus this lens on data, we can improve the quality of data a system creates.

A well-defined SDLC is designed to manage schedule and budget (the project management goals) and to ensure a level of quality for the system that is developed (the development and quality assurance [QA] goal). An SDLC consists of a set of phases, each of which is associated with milestones that mark the progress of the work and deliverables that are work products (designs, code, test cases, and test results) or are used to manage the delivery of work products (project plans). The process is governed by a series of gates—or checkpoints—at which deliverables are reviewed to ensure that they are meeting project goals. The phases themselves may be called by different names, depending on the methodology, but they represent, essentially, a beginning, middle, and end of a project.

A comprehensive SDLC accounts for the true beginning of the system—its conception and feasibility—and the end of a system—its maintenance and obsolescence (Fig. 45). However, because

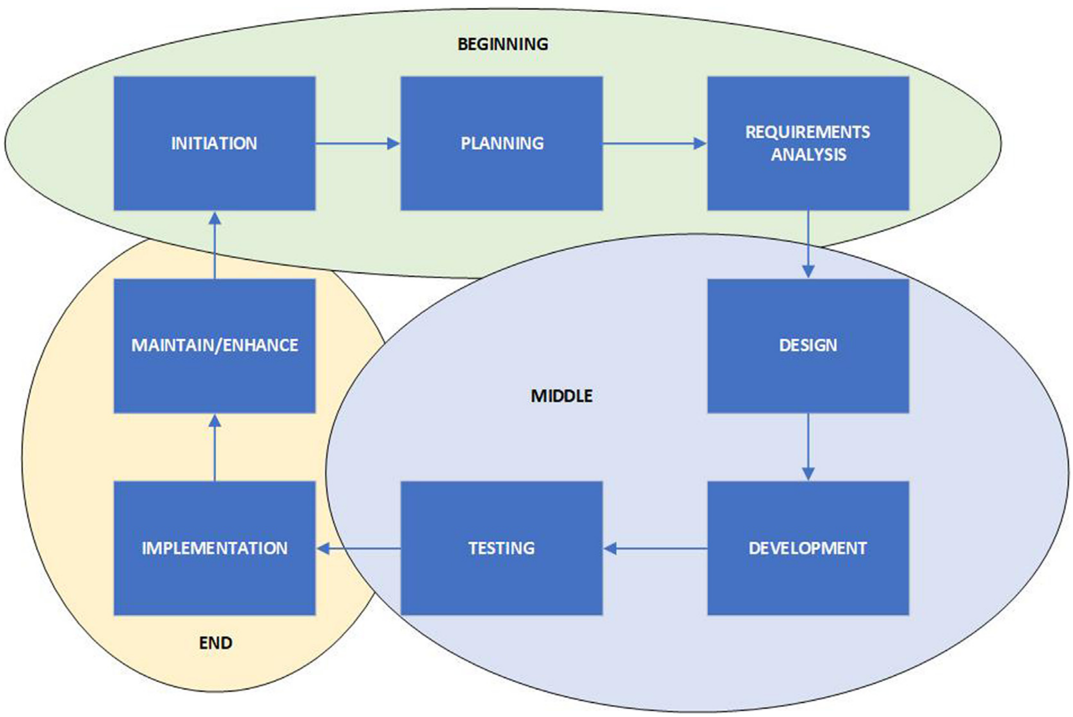

FIGURE 45

The fundamental narrative structure of the systems development life cycle.

the usual use of an SDLC is to manage project activities and deliverables, these end points are often off-stage for most people developing a system. The project has been conceived before development work starts. The project team disbands, and an operational team takes over when a system is put into a production environment. The phases most project team members are familiar with look something like this:

- **Initiation:** Getting the project started, getting funding, getting resources
- **Planning:** Determining the project schedule and dependencies
- **Requirements analysis and definition:** Clarifying and documenting the business needs that the project must fulfill
- **Design:** Defining the solution that will meet the business needs
- **Development:** Building the solution, assembling and configuring the hardware, writing the software
- **Testing:** Ensuring that the functionality described in the specification works the way the specification says it should
- **Implementation:** Putting the hardware and software into an environment where it can be accessed and used by end users, and confirming that the system has been implemented correctly
- **Maintenance:** Handing off the work to a production support team who will be responsible for ensuring the continued functioning of the system

The beginning (initiation, planning, requirements analysis) and middle (design, development, and testing) phases closely resemble the early phases of the asset/resource life cycle. The end phase—implementation—makes the data asset ready for use.[6] Based on this, it is easy to imagine accounting for data as part of the SDLC. Unfortunately, many IT projects focus only on the system and its functionality, and they do not account for the data itself.[7] This failure can result in building technology that simply moves bad data around the organization or that turns good data into bad data and then moves it around the organization.

An example may illustrate the point. When I first started working in IT, as a data quality analyst in an enterprise data warehouse, the data quality section of the specification template included a disclaimer along these lines: "The quality of the data in the warehouse depends on the quality of the source data." This was just a polite way of saying, "We have no control over the quality of data here. 'Garbage In/Garbage Out.'" However, a review of help desk tickets showed that more than three fourths of the issues reported by end users of the warehouse were not rooted in source system data errors. Many were, instead, connected to the ways that data had been stored or maintained in the warehouse. In some cases, this meant that the technical decisions by the development team had created undesired characteristics in the data. The root cause of some of these issues was that the implementation team did not under-

[6]I put this in terms of beginning, middle, and end to make my point that there is a narrative implicit in any SDLC. Someone will probably scratch his or her head and ask: What about Agile? To which I would say (though it may sound like anathema to compare Agile to other methodologies), "Agile is an SDLC, with a beginning, middle, and end. Each program increment is a mini-SDLC (plan, design, develop, test, release, repeat) as is each sprint."

[7]This assertion is based on analysis of the mention of data in the PMBOK *(Project Management Body of Knowledge)*. Thanks for Bob Furce for this analysis.

stand or account for differences in data from different sources when that data was integrated in the warehouse. The data was fine from the perspective of the different source systems, but because the data was structured differently in the sources, it could not be fully integrated unless the warehouse development team recognized these differences when it integrated the data. Instead of "Garbage In/Garbage Out," we had a case of "Pretty Good Data In/Garbage Out."

This is not to imply that software development ignores quality. It does not. The principles of total quality management have been applied to software development itself because software is a product that can have different levels of quality and because "Information systems are a necessary component of the feedback loop in managing an enterprise" (Joshik, n.d.). Software has recognizable and recognized quality characteristics (e.g., usability, reliability, modifiability). Also, the process for developing software often includes techniques focused on ensuring quality as defined by the customer/end user. For example, in traditional software development, customer input can be obtained through joint application development (JAD) sessions. In Agile software development, demos of functionality allow stakeholders to provide feedback on what is being developed before it is completely developed, so the Agile team can respond and better meet requirements. The QA function within the SDLC is focused on ensuring that software meets defined specifications and requirements.

However, the requirements themselves do not specifically focus on the quality of the data being presented to end users by the system. They describe what the system will do. Requirements include what inputs the system will need and what outputs it will produce (both of which are data) as well as how data may need to be stored and maintained in the system and what volume of data the system will be able to process. Some SDLC descriptions even include data analysis and the documentation of data flow within the system, but none directly acknowledge the possibility that the data may not be available or the need to define quality requirements for the data itself.

As discussed in Chapter 6, there is some basis for IT disowning responsibility for data. Making data reliable depends first and foremost on well-defined, reliably executed business processes. In this sense, "the business" should "own" the quality of the data. However, most business processes are executed via IT applications, so these too should be designed to produce reliable data. Also, because data moves through the organization through IT systems, each of which leaves its imprint on the data, IT has a responsibility to ensure that data moves correctly and completely and that it is usable by consumers along the data chain. Doing so requires understanding data structure, identifying and managing data-related risks, implementing controls to prevent damaging or losing the data, and monitoring data in order to inform data consumers of any known limitations of the data. These activities themselves depend on specifying data quality requirements as part of the development or enhancement of a system.

In fact, most SDLC methodologies hardly acknowledge that the purpose of any information system is the information itself. If they did, a lot of the pain and suffering associated with IT development projects would be reduced (Keizer, 2004; Snaplogic, 2021). Given the depth of experience most organizations have with data not meeting expectations, it is hard to sympathize when project teams choose to assume that the data must be okay, rather than checking to see what it looks like. This observation raises the question: What would it look like if data were not only accounted for during development work, but if it were viewed as

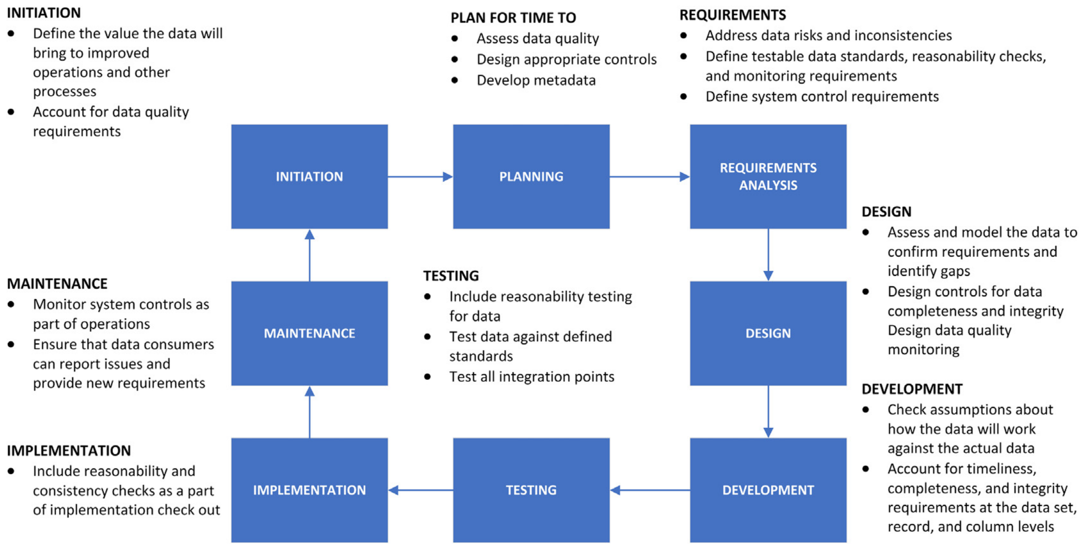

FIGURE 46

Building data quality processes into the SDLC.

the most critical part of the system being developed, the end product of the development cycle? (See Fig. 46.)

- **Initiation**
  - Define the value of the project in terms of the value the data will bring to improved operations and other processes.
  - Identify resources who can define data quality requirements.
- **Planning:** Account for the time needed to
  - Assess data quality and manage any identified risks
  - Design and put in place appropriate data quality controls
  - Develop and groom semantic metadata required to use the data
- **Requirements analysis**
  - Resolve conflicts identified through data risks, including unexpected granularity, invalid values, unknown or inconsistently applied business rules, and unclear or inadequate metadata.
  - Define testable data standards, reasonability and consistency checks, and monitoring requirements.
- **Design**
  - Assess and model the data to confirm requirements and identify gaps.
  - Design controls that will ensure data completeness and integrity and alert production support teams to unexpected system events that affect the data.
  - Design data quality monitoring functionality that will alert data consumers to unexpected characteristics of the data.

- **Development**
  - Check assumptions about data against the actual data.
  - Account for timeliness, completeness, and integrity at the data set, record, and column levels.
- **Testing**
  - Include reasonability testing for data, in addition to testing system functionality
  - Test data against defined standards.
  - Test the completeness of all integration points.
- **Implementation**
  - Include data reasonability and consistency checks a part of the implementation check out.
- **Maintenance**
  - Monitor system controls as part of operations.
  - Ensure that data consumers can get their questions answered, report issues, and provide feedback, including new requirements.

Concluding Thoughts

Data has characteristics of a resource or asset that organizations use to generate value. In this way, data can be understood as similar to equipment, financial resources, or even human resources (people with particular knowledge and skills needed by the organization). If an organization is looking to improve overall data management and implement data quality by design, it helps to look at data from the perspective of the data life cycle and identify the points in that life cycle where it can define its quality requirements. The supply chain, as a means of understanding the movement of data within an organization, presents the opportunity to clarify expectations for the quality of data supplied to a range of organizational processes and to provide feedback if the data does not meet these requirements. The value chain allows us to ask questions about the uses of data, the costs associated with it, and the ways it may bring value.

Importantly, value chain analysis can be used to stop costly processes if they are not adding value. These two complementary models work best when organizations are looking at their products or services. The idea of the data supply chain shows features of how data is created by data producers and used by data consumers within a particular organization or in support of a specific goal. It is useful to know these details, whether one is trying to improve the organization's processes or address a specific data quality issue, because they remind us that we can produce higher-quality data by controlling the processes that create data.

Ultimately, data creation and development processes are implemented through projects. Including data quality management components as part of the SDLC improves the chances of delivering and maintaining reliable data.

These multiple models of data management show us the ways in which data is integral to all of an organization's activities. Data binds organizations together in complex ways. In revealing these connections and complexities, the models also show us a way forward. There are many opportunities to improve the quality of organizational data. Being aware of these is the first step to making better data. Purposefully connecting them to core data quality and data management practices is the next step.